

👉 XAI Lecture 11

Counterfactual Explanations & Algorithmic Recourse

FAMAF - UNC

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eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

🍷 Paper 1 ---



Introduction + Motivation ---

****EU General Data Protection Regulation (2018)****

- "the toughest privacy and security law in the world" - Article 13-14, regarding automated decision-making: "meaningful information about the logic involved" - Recital 71: "the right to obtain an explanation of the decision reached and to challenge the decision"

4 Key Problems

1. Not legally binding
2. Only applicable in limited cases
3. Explainability is technically very challenging
4. Competing interests of data controllers, subjects, etc.

🍷 Unconditional Counterfactual Explanations ---

Authors propose

- **3 aims for explanations** - Inform and help the subject understand "why" - Provide grounds to contest decisions - Understand options for recourse - **CFEs** - Fulfill the above goals - Overcome challenges w.r.t. current interpretability work - Can bridge the gap between interests of data subjects and data controllers

Do you buy this?

🍷 Unconditional Counterfactual Explanations ---

- **Definition:** - A statement of how the world would have to be different for a desirable outcome to occur - **Because** of features $\{x_1, x_2\}$, x 's outcome was label y_1 - **If**
 $x = \{x_1 + \delta_1, x_2 + \delta_2\}$, **then** y_2 . -
Example: - "You were denied a loan because your annual income was \$30000. If your income has been \$45000, you would have been offered a loan"

Notes

- No one **unique** CF - Need to consider **actionability** (mutability of variables) - Providing **several diverse** CFEs may be more useful than simply the closest/shortest one

🍷 Background + Related Work ---

- Historic context of knowledge
- Prior explainability work
- Adversarial examples/perturbations
- Causality/Fairness

🍷 Background: Historic Context of Knowledge ---

- In order to know something, it is not enough to simply ****believe**** that it is true: rather, you must also have a good ****reason**** for believing it
- If q were false, S would not believe p

Note

- This statement only describes S 's **beliefs**, which might not reflect reality - This statement can be made without knowledge of the **causal relationship** between p and q - Q: who is S ? The user? The model? Us?

Background: Previous Explanations in AI/ML ---

- Previously: providing insight into the internal state of an algorithm, human-understandable approximations of the algorithm
- Three-way tradeoff between:
 - ① quality of approximation
 - ② ease of understanding the function
 - ③ size of the domain for which the approximation is valid
- **CFEs:**
 - ▶ Minimal amount of information
 - ▶ Require no understanding of internal logic of model
 - ▶ No approximation (although might not always be minimum length)
 - ▶ Con: May be overly restrictive

🍷 Background: Adversarial Perturbations ---

- Small changes to a given image can result in the image being assigned to a different class
- Very similar to CFEs but the changes aren't necessarily sparse
- Often not human perceptible
 - ▶ Authors propose that this is because the new images lie outside the “real-image” manifold
 - ▶ Emphasize the importance of solutions/CFEs being *possible* as well as close
 - ▶ Further research into structure of high-D data is structured before CFEs can be useful/reliable in those domains

🍷 Background: Causality and Fairness ---

- Can provide evidence that models/decisions are affected by protected attributes - If CFEs change one's race, the treatment of that individual is dependent on race

Discussion

Is the converse true?

Summary of Contributions ---

- Highlights the difficulties with conveying the inner workings of modern ML algorithms to users
 - ▶ Complexity
 - ▶ Lack of utility (except for “builders”)
- Introduces an algorithmic approach to counterfactuals
 - ▶ Rooted in adversarial machine learning
- Connects counterfactuals to the GDPR
 - ▶ Demonstrates advantages over other interpretability methods from a policy perspective



Approach: Counterfactual Optimization ---

$$\arg \min_{x'} \max_{\lambda} \lambda (f_w(x') - y')^2 + d(x_i, x')$$

- **Minimize:**

- ▶ Squared error with desired (counterfactual) label
- ▶ Distance to perturbed point

- **While maximizing** weight on squared error term

- ▶ We want to have the prediction change more than we want the point to be close
- ▶ Iteratively: solve for x' , then maximize λ

- **Settings:**

- ▶ Tabular Data
- ▶ Regression

Approach: Distance Metrics ---

$$d(x, x') = \sum_{k \in F} \frac{\|x_{i,k} - x'_k\|_p}{N}$$

****Norms****

$$|x_{i,k} - x'_k|$$

$$(x_{i,k} - x'_k)^2$$

****Normalizing Factors****

$$MAD_k := \text{median}_{j \in P} (|X_{j,k} - \text{median}_{l \in P}(X_{l,k})|)$$

$$\text{std}_{j \in P}(x_{j,k})$$

🍷 Experimental Results: LSAT ---

$$d(x_i, x') \propto \sum_{k \in F} (x_{i,k} - x'_k)^2$$



🍷 Experimental Results: LSAT (L1) ---

$$d(x_i, x') = \sum_{k \in F} \frac{|x_{i,k} - x'_k|}{MAD_k}$$



Experimental Results: Discussion ---

- **Realistic (or achievable) counterfactual values**
 - ▶ Clipping and Clamping
- **Counterfactual Targets**
 - ▶ LSAT: Avg. Grade 0
 - ★ Students would want to know how to improve their average grade
 - ▶ PIMA: Risk Score of 0.5
 - ★ Patients would want to know how to achieve a lower risk score

🍷 Explanations and The GDPR ---

****GDPR Requirements: Recital 71****

- Implement suitable safeguards against automated decision-making - Include specific information to the data subject and the right to obtain human intervention - To express their point of view - To challenge the decision - To obtain an **explanation** of the decision reached after such assessment

- Does not require opening the "black box" to explain the internal logic of decision-making systems - Does not explicitly define requirements for explanations of automated decision-making

🍷 Counterfactual Explanations and The GDPR ---

- Legislators wanted to clarify that some type of explanation can voluntarily be offered after a decision has been made
- Many aims for explanations are feasible
- Emphasis of GDPR is on protections and rights for individuals

🍷 Advantages of Counterfactual Explanations ---

- Bypass explaining the *internal workings* of complex machine learning system
- Simple to compute and convey
- Provide information that is both easily digestible and practically useful
 - ▶ Understanding the reasons for a decision
 - ▶ Contesting decisions
 - ▶ Altering future behaviour to receive a preferred outcome

Possible mechanism to meet the explicit requirements and background aims of the GDPR

Broader Possibilities with the Right of Access ---

Art 15.

- Confirm whether or not personal data used
- Provide information available after a decision has been made
- Avoid disclosing personal data of other data subjects
- Balance interest of the subject and controller
 - ▶ Potential to contravene trade secrets or intellectual property rights

Counterfactuals

- Data of other data subjects or detailed information about the algorithm does not need to be disclosed - Disclose only the influence of select external facts and variables on a specific decision - Less likely to infringe on trade secrets or privacy

🍷 Understanding Through Counterfactuals ---

Art 12(1)

- Requires information to be conveyed in a “concise, transparent, intelligible and easily accessible form”
- CFEs align with this requirement by providing simple “if-then” statements
- CFEs provide greater insight into the data subject's personal situation as opposed to an overview tailored to a general audience

Minimally burdensome and disruptive technique to understand the rationale of specific decisions

🍷 Legal Information Gaps on Contesting Decisions ---

- **Art. 16**

- ▶ Data subject has the right to correct inaccurate data used to make a decision, but does not need to be informed of which data the decision depended

- **Art. 22**

- ▶ Data subjects do not need to be informed of their right *not* to be subject to an automated decision

🍷 Contesting Through Counterfactuals ---

- Lead to greater protection for the data subject than currently envisioned by the GDPR
- Align with Article 29 Working Party
 - ▶ Understanding decisions + knowing legal basis is essential for contesting decisions
- Reduce burden on data subject
 - ▶ Understand most influential data (instead of vetting all collected data)
 - ▶ Compact way to convey dependencies

Explanations to Alter Future Decisions ---

- **GDPR**

- ▶ Explanations not explicitly mentioned as a guide to altering behavior to receive a desired automated decision

- **Article 29 Working Party**

- ▶ Provide suggestions on how to improve habits and receive better outcome

- **Counterfactual explanations**

- ▶ Can address the impact of changes to more than one variable on a model's output at the same time
- ▶ Can be used in a contractual agreement between data controllers and data subjects

🍷 Discussion ---

- **Who do these explanations serve?** - Developers, Data subjects, and Lawmakers -
- Do you think they are misinterpreted?** -
- Causality - Optimizing for sparsity may not reflect reality - **Priors and desired outcomes**

(Yacoby et al. 2020)



🍷 Paper 2 ---



🍷 Algorithmic Recourse ---

- Increasingly algorithms are used to make consequential decisions for individuals - Recourse: "**Systematic process of reversing unfavorable decisions made by algorithms and bureaucracies**"
- Promoting **agency** and **trust**



Paper's Main Contributions ---

- ❶ **Insufficiencies of previous problem formulations**
 - ▶ → Motivates causal approach to recourse
- ❷ **Structural Causal Model approach to recourse**
- ❸ **Discuss examples motivated by real-world problems and future directions**

🍷 Nearest Counterfactual Explanations ---

- For a person with features $\mathbf{x}^F$ who was denied a loan, find the "nearest neighbor" $\mathbf{x}$ who was granted a loan - Difference between $\mathbf{x}^F$ and $\mathbf{x}$ is an "explanation" for the loan denial for $\mathbf{x}^F$



🍷 Nearest Counterfactual Explanation (CFE) ---

$$\mathbf{x}^{*CFE} \in \arg \min_{\mathbf{x}} \text{dist}(\mathbf{x}, \mathbf{x}^F) \quad \text{s.t.} \quad h(\mathbf{x}) \neq h(\mathbf{x}^F), \mathbf{x} \in \mathcal{P}$$

- **Finding nearest counterfactual explanation as an optimization problem**
 - ▶ $\mathbf{x}^{*CFE}$ is the Counter Factual Explanation
 - ▶ Function h is the classifier
- **Distance metrics**
 - ▶ Lp norm; L1 norm divided by median absolute deviation; etc.

🍷 Why Are CFEs Not Ideal for Recourse? ---

Counterfactual Explanations provide **understanding** but do not necessarily lead to optimal **action recommendations**

🍷 Why Are CFEs Not Ideal for Recourse? ---

- ① Don't account for person's difficulty or “cost” of changing dimensions of $\mathbf{x}^F$ (addressed by Ustun et al.)
- ② Don't account for downstream “causal” impact of taking actions (addressed by this work)

🍷 Accounting for the "Cost" of Actions (Ustun et al.) ---

$$\delta^* \in \arg \min_{\delta} \text{cost}(\delta; \mathbf{x}^F) \quad \text{s.t.} \quad h(\mathbf{x}^{\text{CFE}}) \neq h(\mathbf{x}^F),$$
$$\mathbf{x}^{\text{CFE}} = \mathbf{x}^F + \delta, \quad \mathbf{x}^{\text{CFE}} \in \mathcal{P}, \quad \delta \in \mathcal{F}$$

- δ^* restricted to the set of “feasible changes”
- Consider linear impact of changes δ
- Non-trivial to choose costs that reflect people’s true objective functions

(Ustun, Spangher, and Liu 2019)



Insufficiencies of Ustun et al. Formulation ---

$$\delta^* \in \arg \min_{\delta} \text{cost}(\delta; \mathbf{x}^F) \quad \text{s.t.} \quad h(\mathbf{x}^{\text{CFE}}) \neq h(\mathbf{x}^F)$$

① Marginal cost of changing a feature is constant

- ▶ **Example:** Cost of going from salary \$0 to \$100k equals cost to go from salary \$800k to \$900k

② Doesn't consider downstream “causal” impact of actions

- ▶ **Example:** Loan decision for individual changed if “salary” reaches 100k (+33%) or “bank balance” reaches 30k (+20%). 20% change seems easier.
- ▶ However, best **action recommendation** is to increase salary by 14% when 30% of salary automatically saved to bank.

🍷 Actions as Interventions: Structural Causal Model (SCM) ---

Setup

- $M (M \in \Pi) = \langle F, X, U \rangle$: SCM - X : endogenous (observed) variables - U : exogenous (unobserved) variables - F : $U \rightarrow X$, structural equations - $A (\Pi \rightarrow \Pi)$: structural interventions, i.e. transformations between SCMs - of the form $A := \text{do}(\{X_i := a_i\}_{i \in I})$



- M : true world model - observation:
 $M_1 \neq M_2 \neq M_3$

Actions as Interventions: Structural Counterfactuals ---

What will individual x^F 's feature vector be after the individual performs action set A in world M ?

Assumptions: (1) no hidden confounders (true SCM), (2) full access to invertible F

Idea: once we know F , X (endogenous variables) can be uniquely determined given U (exogenous variables)

Compute $F^{-1}(x^F)$

Takeaway: we can compute any structural counterfactual query for individual x^F :

$$x^{\text{SCF}} = F_A(F^{-1}(x^F))$$

🍷 Limitations of CFE-Based Recourse: Formalism ---

Setup: x^F (individual features), δ^* action recommendation (Ustun et al. solution), I (set of indices of acted-upon observed variables: $I = \{i \mid \delta_i^* \neq 0\}$)

Definition (CFE-Based Actions): a set of structural interventions
 $A^{\text{CFE}} := \text{do}(\{X_i := x_i^F + \delta_i^*\}_{i \in I})$

Proposition: $A^{\text{CFE}} \rightarrow x^{\text{SCF}} = x^{*\text{CFE}} := x^F + \delta^*$ (i.e. recourse is guaranteed) **if and only if** I 's descendants = $\{\}$.

Corollary: if the true world M is independent—if all the observed features are root-nodes of G —then CFE-based actions always guarantee recourse.

Algorithmic Recourse via Minimal Interventions: Setup ---

Formulation

$$A^* \in \arg \min_A \text{cost}(A; \mathbf{x}^F)$$

$$\text{s.t. } h(\mathbf{x}^{\text{SCF}}) \neq h(\mathbf{x}^F)$$

$$\mathbf{x}^{\text{SCF}} = \mathbb{F}_A(\mathbb{F}^{-1}(\mathbf{x}^F))$$

$$\mathbf{x}^{\text{SCF}} \in \mathcal{P}, \quad A \in \mathcal{F}$$

Remarks

- finding minimal shift of features $\rightarrow$ finding minimal cost action set that yields favorable label - $A^* \in \mathcal{F}$ = set of feasible actions with minimally costly recourse - $\text{cost}(\square; \mathbf{x}^F): \mathcal{F} \times \mathcal{X} \rightarrow \mathbb{R}_+$ - $\square^{\text{SCF}} \neq \square^{\text{CFE}}$ (from Ustun et al.)!

Algorithmic Recourse via Minimal Interventions: Formalism ---

$$A^* \in \arg \min_A \text{cost}(A; x^F) \quad \text{s.t.} \quad h(x^{\text{SCF}}) \neq h(x^F), \quad x^{\text{SCF}} = F_A(F^{-1}(x^F)), \quad x^{\text{SCF}} \in \mathcal{P}, \quad A \in \mathcal{F}$$

Proposition

- A^{CFE} : Counter Factual Explanation-based action - A^* : Minimal Intervention Solution

$$\text{cost}(A^*; x^F) \leq \text{cost}(A^{\text{CFE}}; x^F)$$

🍷 Algorithmic Recourse via Minimal Interventions: MINT ---

- **Recourse through Minimal Interventions (MINT) idea:**

- ▶ Required: that we can compute structural counterfactual of an individual in the world given *any* feasible action
- ▶ Focus on the case where the SCM is an additive noise model
- ▶ $\Rightarrow$ Abduction-action-prediction technique (Pearl et al.) to compute x^{SCF} : $F_A(F^{-1}(x^F))$

🍷 Abduction-Action-Prediction to Obtain x^{SCF} ---



$\{U_i\}_{i=1}^4$: mutually independent, exogenous
 $\{f_i\}_{i=1}^4$: structural equations
 $x = [x_1^F, x_2^F, x_3^F, x_4^F]^T$: observed factual features

1. ****Abduction:**** compute exogenous variables -
 $u_1 = x_1^F, u_2 = x_2^F, u_3 = x_3^F - f_3(x_1^F, x_2^F),$
 $u_4 = x_4^F - f_4(x_3^F)$
2. ****Action:**** modify SCM with interventions -
 $X_1 := [1 \in I] \cdot a_1 + [1 \notin I] \cdot U_1 -$
 $X_2 := [2 \in I] \cdot a_2 + [2 \notin I] \cdot U_2$
3. ****Prediction:**** recursively compute endogenous variables -
 $x_1^{\text{SCF}} := [1 \in I] \cdot a_1 + [1 \notin I] \cdot u_1 -$
 $x_3^{\text{SCF}} := [3 \in I] \cdot a_3 + [3 \notin I] \cdot (f_3(x_1^{\text{SCF}}, x_2^{\text{SCF}}) + u_3)$

General Formulation and Solving the Optimization Problem ---

General Formulation

$$A^* \in \arg \min_A \text{cost}(A; \mathbf{x}^F)$$

$$\text{s.t. } h(\mathbf{x}^{\text{SCF}}) \neq h(\mathbf{x}^F)$$

$$x_i^{\text{SCF}} = [i \in I] \cdot (x_i^F + \delta_i)$$

$$+ [i \notin I] \cdot (x_i^F + f_i(\mathbf{pa}_i^{\text{SCF}}) - f_i(\mathbf{pa}_i^F))$$

$$\mathbf{x}^{\text{SCF}} \in \mathcal{P}, \quad A \in \mathcal{F}$$

****Remarks****

- $x_i^F + \delta_i$: intervention - $f_i(\mathbf{pa}_i^F)$: factual values of x_i 's parents - $f_i(\mathbf{pa}_i^{\text{SCF}})$: counterfactual values of x_i 's parents - new closed-form expression for $F_{A^*}(F^{-1}(x^F)) \rightarrow$ use optimization methods

Experimental Setup ---

Synthetic setting:

Generate data following causal generative process

Real-world setting:

Use existing German credit dataset to learn structural causal model equations, by fitting a linear regression

Cost for both is ℓ_1 norm over normalized feature change

🍷 Experimental Setup: Synthetic Setting ---



Causal Generative Process

$$U_1 \sim \$10000 \cdot \text{Poisson}(10), \quad U_2 \sim \$2500 \cdot N(0, 1)$$

$$X_1 := U_1$$

$$X_2 := f_2(X_1) + U_2, \quad X_2 := \frac{3}{10} \cdot X_1 + U_2$$

$$\hat{Y} = h(X_1, X_2), \quad h = \text{sgn}(X_1 + 5 \cdot X_2 - \$225000)$$

🍷 Experimental Results: Synthetic Setting ---

[annual salary, bank balance]



🍷 Experimental Results: Synthetic Setting (cont.) ---



$\mathbf{x}^{\text{SCF}}$ further dist from $\mathbf{x}^F$ than $\mathbf{x}^{\text{CFE}}$

BUT

$$\text{cost}(\delta^*; \mathbf{x}^F) \approx 2 \text{cost}(A^*; \mathbf{x}^F)$$

🍷 Experimental Setup: Real-World Setting ---



Structural Causal Model

$$X_1 := U_1, \quad X_2 := U_2$$

$$X_3 := f_3(X_1, X_2) + U_3$$

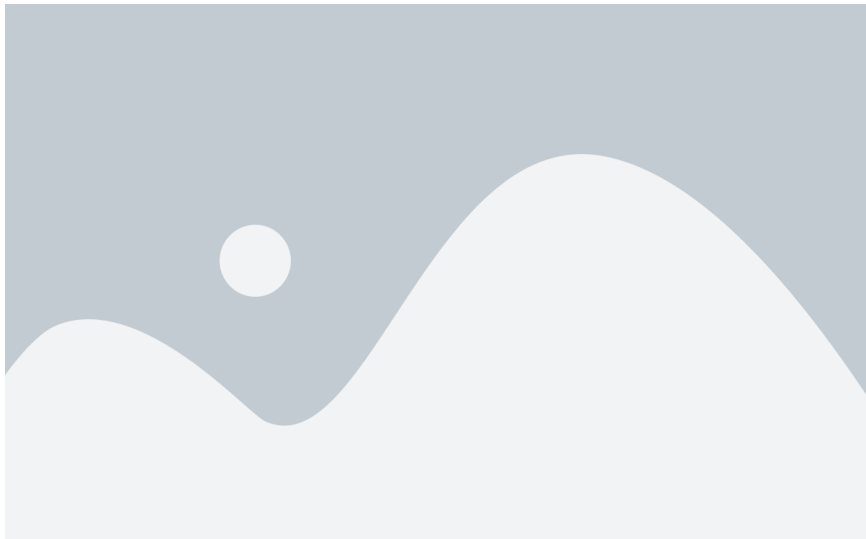
$$X_4 := f_4(X_3) + U_4$$

$$\hat{Y} = h(\{X_i\}_{i=1}^4)$$

h can be logistic regression or decision tree

🍷 Experimental Results: Real-World Setting ---

[gender, age, credit given, credit repayment duration]



🍷 Experimental Results: Real-World Setting (cont.) ---



42% decrease in cost using Karimi et al.'s formulation

Averaged over 50 test individuals, $39 \pm 24\%$ and $65 \pm 8\%$ decrease in cost, for h as logistic regression and decision tree, respectively

Future Work: Extended Kinds of Interventions ---

****Forms****

- ****Structural/hard**** (actions in Karimi et al.): unconditionally sever all edges incident on intervened node -
- **Additive/soft:**** do not sever incident edges

$$x_i^{\text{SCF}} = [i \in I] \cdot \delta_i + (x_i^F + f_i(\mathbf{pa}_i^{\text{SCF}}) - f_i(\mathbf{pa}_i^F))$$

****Scopes****

- Karimi et al. assumes action = intervention on endogenous variable -
- **Fat-hand/non-atomic:**** confounded/correlated interventions

****Feasibility****

Can encode as constraints to amend to $A \in \mathcal{F}$

- ****Immutable:**** closed under ancestral relationships - ****Mutable but non-actionable:**** $[i \notin I] = 1$ is sufficient - ****Actionable and mutable:**** contingent on (a) pre-intervention value of variable (b) pre-intervention value of other variables (c) post-intervention value of variable (d) post-intervention value of other variables

🍷 Future Work: Current Limitations ---



- **Reliance on true causal model of the world**

🍷 Concluding Thoughts ---

- Overall an interesting work bridging counterfactual explanations (previous papers) and algorithmic recourse (next papers), clarifying their differences
- Tradeoff between generalizability of setting and hardness of optimization problem
 - ▶ Ustun et al. pursue algorithmic recourse in a specific linear setting
 - ▶ Karimi et al. pursue algorithmic recourse in general settings, require true causal model of world
 - ▶ Open question: where do we stand on this tradeoff? Consider the origins of counterfactual explanations in Wachter et al.

Thank You!

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